Zachary Burton

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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA
B.S. in Mathematics (GPA: 4.6/5.0) *Sep 2022 – Feb 2026*

Coursework: Probability, Statistics, Real Analysis, Linear Algebra, Abstract Algebra, Reinforcement Learning (Grad), Program Synthesis (Grad)

Publications

QEDBENCH: Quantifying the Alignment Gap in Automated Evaluation of University-Level Mathematical Proofs | Accepted at ICML 2026, Co-authorship

Structured Hints for Sample-Efficient Lean Theorem Proving | arXiv:2601.16172

Poissonization-based collision threshold derivation for random walks on lattices | arXiv:2505.02973

Experience

Meta Feb 2026 – Present
Machine Learning Engineer *New York, NY*

- Shipped a revenue optimization launch boosting billing-based revenue by **+0.251%** across Facebook Reels surfaces; fastest ramp-to-launch on team
- Designed and drove a cohort-aware calibration system for a listwise ranking model, segmenting users into decile-based revenue cohorts and iterating through 7 experiment configurations with a **+0.22%** revenue lift in early signal
- Co-authored two launch proposals for position-based video length tuning in a late-stage reranker, yielding **+0.185%** revenue-per-view (head-load) and **+0.525%** tail-load watch time globally

Software Engineer Intern *May 2025 – Aug 2025*

- Refactored copyright enforcement engine processing **2B+** daily uploads; achieved **97%** consistency with legacy outputs, verified via SQL logging and automated regression suite
- Uncovered and fixed **16%** false-positive error affecting 3 internal teams; designed data quality checks in Hack/PHP to detect and surface annotation inconsistencies

Independent Research

Structured Hints for Neural Theorem Proving 2026
Independent project | arXiv:2601.16172

- Showed that RL-trained provers (DeepSeek-Prover-V1.5) underutilize structural priors in the Lean 4 tactic language; designed a fixed prompt schedule over 15 common tactic skeletons as inference-time guidance
- Achieved **21.7%** pass@16 on miniF2F vs. **15.2%** baseline (43% relative improvement) with no retraining, same sample budget (k=16), and same max generation length (1024 tokens)

RL Agents for Symbolic Algebra Fall 2025
MIT 6.7920 | PyTorch, PPO, custom algebra env

- Designed custom environment mapping algebraic equation solving to sequential decision-making over AST state spaces with $\sim$20 symbolic rewrite rules and sparse rewards
- Tested OOD generalization for PPO, Q-Learning, and Sarsa on unseen linear, quadratic, and recurrence tasks, showing curriculum benefits for symbolic reasoning depth

RL for Automating NL-Completeness Reductions Nov 2024 – Aug 2025
MIT CSAIL | Advised by Jayson Lynch & Erik Demaine

- Formulated gadget simulation search—automating motion planning NL-completeness reductions—as an MDP over finite-automaton gadget structures with $\sim 10^4$ discrete actions; Maskable PPO agent achieves **85%** success in <3ms vs. 2% random and >100s exhaustive baselines across 1,000 held-out instances
- Identified generalization failure: agents memorize task-specific action traces rather than learning compositional construction logic; catastrophic failure under planarity constraints and held-out tasks